Request for Proposals (RFP)

Open-H-Hospital: Open data initiative for hospital workflow robotics

1. Introduction

Hospitals face growing operational complexity, workforce constraints, and increasing demand for efficient, high-quality care. Robotics and Physical AI offer the potential to automate repetitive, labor-intensive, and physically demanding tasks across hospital environments. However, progress is constrained by fragmented data, limited access to real-world demonstrations, and the lack of open, reusable models and evaluation frameworks.

Open-H-Hospital is a collaborative effort led by NVIDIA and partner institutions to build open foundations for robotics-mediated hospital workflows. The effort centers on two distinct tracks distinguished by data type. A team may take part through either or both:

- **Track A: Hospital robotics datasets.** Data collected from hospital robots in-situ. This can include robots of any type and with any purpose within the hospital setting including supplies stocking, sample conveyance or delivery, medication dispensing, procedural, or other.
- **Track B: Hospital workflow datasets.** Annotated video data that informs flow of patients, hospital staff, equipment, or supplies in or around a hospital, either in a specific room or bay (for example, the operating room, walk-in clinic) or these datasets can optionally also be accompanied by related structured data that speak to hospital or department performance metrics, for example, patient length of stay, patient outcome, procedure duration, procedure outcome, length of stay, or other.

The primary scope of Open-H-Hospital concerns datasets that inform workflows in, through, and around hospital spaces. Datasets that focus on the surgical field in laparoscopic robotic surgery or endoluminal surgery fall within the scope of the sibling Open-H initiatives: Open-H-Embodiment and Open-H-Endoluminal initiative, and should instead be submitted at github.com/open-h/open-h-endoluminal respectively. The key distinction is that Open-H Embodiment and Endoluminal focus on surgical robotics and surgical video/data (within the body), while Open-H Hospital expands the mission to exterior spaces within the hospitals for hospital-wide robotic automation—including logistics, nursing support, environmental services, and general-purpose mobile manipulation.

Contributions are accepted across multiple tiers of realism, including clinical, staged, and simulation.

2. Steering Group

Role	Name	Affiliation
Industry Lead	Dr. Mostafa Toloui	NVIDIA
Academic Lead	Dr. Michael Yip	Cedars Sinai
Clinical Lead	TBD	TBD

3. Objectives and Outcomes

1. **Create the foundation**. Build the first and largest open dataset for hospital workflows.
2. **Commercially usable models.** Use the annotated data to train commercially usable WFMs, VLMs, VLAs, WAMs, and CV expert models.
3. **Open release.** Publish cleaned data, annotations, and model artifacts under a permissive license.
4. **Weak, scalable labeling.** Develop and apply a weak agentic labeling toolkit that propagates a seed of expert annotations to much larger datasets, so annotation scales beyond what manual labeling alone can reach.

4. Scope of Work

Participation is organized into two tracks. A team may propose to either or both. Defined below are acceptable track contributions and requirements for both tracks.

4.1 Track A: Healthcare Robotics Data

Contribute synchronized recordings from one or more of the following groups.

Healthcare Robotics. Examples: mobile radiology robots; X-ray positioning; autonomous mobile robots for delivering medications, supplies, and lab specimens; humanoids and mobile manipulators for restocking, material handling, and room turnover; robotic nursing assistants for patient transport and bedside support; and cleaning and service robots for disinfection, waste handling, and hospital logistics. **Platforms**: mobile manipulators, collaborative arms, autonomous mobile robots (AMRs), humanoids, robotic nursing assistants, delivery robots, cleaning robots, and specialized service robots for hospital logistics and operations.

4.3 Data Requirements and Contribution Tiers (Track A)

Each data proposal must commit to a minimum hour contribution and to providing time-aligned streams and required metadata. Proposals should include the following data types

- **Time-aligned streams.** Provide, synchronized: camera streams (RGB or RGB-D, wrist and room); robot kinematics (pose, force and torque, and other state); and corresponding medical images or video (for example ultrasound B-mode or surgical video).
- **Kinematics.** Native Cartesian end effector kinematics and optionally joint-space kinematics are required for all submitted datasets. However, on a case-by-case basis video-only datasets with inferred poses that demonstrate a high level of accuracy will be considered as well.

Additionally, the following data types are welcome but optional.

- *[Optional]* **Tactile**. Force-sensing data, Sensor data capturing contact, pressure, grip, slip, and force during tasks such as handling medical supplies, opening doors, and manipulating carts.
- *[Optional]* **Nonvisual**. Audio, weight distribution, temperature before, during, and after tasks.

Format. Deliver data in the LeRobot data format. Where a robot is involved, supply calibration data and robot CAD and kinematic-tree descriptions (USD, URDF, DH parameters, or equivalent).

Minimum contribution. Each collection setting carries a minimum in hours of synchronized data (below), and a team may contribute across several settings. Higher settings carry a higher relative weight, reflecting their rarity and potential for real-world impact. Preferred order of collection setting: clinical, in-vivo, ex-vivo, phantom or table-top, simulation.

Data Tier	Definition	Minimum Contribution
Real World	Data captured during actual hospital operations, reflecting real workflows, people, environmental variability, and operational constraints. *Examples*: AMR delivering medications between hospital departments; robot navigating crowded corridors and elevators; supply restocking during active clinical operations; human-robot interactions in nursing units.	20 h
Staged	Data collected using physical robots in controlled laboratory or mock-hospital environments, where tasks, objects, layouts, and conditions can be deliberately configured and repeated. *Examples*: humanoid picking up medical supply bins; mobile manipulator loading a delivery cart; robot opening doors, handling trays, or sorting supplies in a mock hospital room.	20 h
Simulation	Data generated in virtual hospital environments using digital twins, physics simulation, synthetic sensors, and procedural scenario generation to scale training and test rare or difficult situations. *Examples*: thousands of simulated delivery runs; robots navigating blocked corridors or crowded elevators; synthetic demonstrations of object manipulation; testing recovery from dropped objects or unexpected obstacles.	60 h

We encourage Track A proposals to also provide time-aligned narration or description of the sub-task (audio or text), correlated anonymized patient information where available, camera intrinsics, and labels for demonstration quality (expert, intermediate, novice). Additionally, task failures and DAgger-style recovery demos are as valuable as successful demos, thus including tasks success flags with each submitted episode is strongly encouraged (failure, recovery, success).

4.2 Track B: Hospital Workflow Data

Contribute hospital workflow data from one or more of the following groups. **Track B data must be accompanied by time-aligned narration or description of the sub-task (audio or text)**. We also encourage submissions to include correlated anonymized patient information where available, camera intrinsics, and labels for demonstration quality (expert, intermediate, novice). Track B proposals can include one or both of the following data types.

- **Exocentric video.** Security footage, tripod, or carried by a third party. Provide RGB or RGB-D camera streams. *Example: bird's-eye video from the ceiling of an operating room while a surgical team performs a surgical procedure.*

- **Egocentric video.** Provide RGB or RGB-D camera streams for camera devices worn by the hospital staff navigating the hospital workflow. *Example: surgeon or surgical technician wearing a video camera during the surgical procedure in the operating room.*

Data Tier	Definition	Minimum Contribution
Clinical	Collected during working hospital hours with appropriate IRB or ethics oversight.	50 h
Staged	Collected outside of hospital hours with people and procedures staged to mimic normal hospital operations.	50 h
Simulation	Digitally simulated using physics-based or differentiable environments.	100 h

4.5 Focused Tracks and Alternative Formats

To capture the benefits of focused, high-quality datasets while keeping the initiative flexible, we invite groups to propose focused tracks around specific areas.

5 Timeline

Milestone	Date
RFP released	October 21, 2026
Proposal submission deadline	December 30, 2026
Data submissions deadline	February 14, 2027
Target preliminary dataset release	March 2027
Model training and validation	March 2027 to June 2027
Target release of final dataset, annotations, and models	June 2027

6. Eligibility and Partnership Models

- Academic institutions, start-ups, and industrial healthcare firms are welcome.
- International participation is encouraged.
- Consortia of multiple organizations may submit a joint proposal with a single lead.

Collaboration benefits. All accepted collaborators will be named co-authors on the Open-H-Hospital dataset publication and on the follow-up paper describing the resulting models. Teams will receive early-access to the Open-H-Hospital evaluation checkpoints of the models and the dataset itself for their own downstream experiments, and recognition during dataset and model releases.

7. Proposal Requirements

Submissions (single PDF) must include:

1. **Executive summary.**
2. **Track and target:** whether you propose to Track A (healthcare robotics), Track B (hospital workflow), or both; for Track A, the target task groups, robots, sensors, collection environment, and committed hours per collection setting; for Track B, the target procedures, annotation types, collection environment, and committed labels per type.
3. **Technical approach:** for data, the collection methodology, synchronization, quality assurance, and privacy safeguards (HIPAA and GDPR); for any included labels, the annotation methodology, tooling, ontology alignment, and quality assurance.
4. **Team qualifications:** key personnel biographies and prior related work.
5. **Project plan:** work-breakdown structure and Gantt chart aligned with the Section 5 timeline.
6. **Data rights and IP statement:** confirmation of intent to release data and annotations under the OpenMDW-1.1 license (or compatible) and to comply with steering-group IP policy.

Submit proposals to the Open-H-Embodiment v2 Google Form.

8. Data Governance, Ethics, and Compliance

- **Patient privacy.** All human-subject data must be de-identified to HIPAA Safe Harbor standards or equivalent (GDPR for European contributors), including scrubbing imaging and video metadata on export.
- **Regulatory.** Proposers are responsible for obtaining IRB or ethics approval where required.
- **Annotation provenance.** Annotations should record who or what produced them (expert, trainee, or model-assisted).
- **Licensing.** Final datasets and models will be released under the OpenMDW-1.1 license. Synthetic assets must be free of third-party IP encumbrances.
- **Release.** Contributors accept that their datasets will be uploaded and released openly via public distribution platform such as HuggingFace.

9. Contact Information

Administrative questions:

- Erin Chu, erchu@nvidia.com (NVIDIA)
- Sean Huver, shuver@nvidia.com (NVIDIA)

Data, annotation, and engineering questions:

- Lukas Zbinden, lzbinden@nvidia.com (NVIDIA)
- TBD from working group